

Strategic Collisions

Can anyone catch the cell phone king?

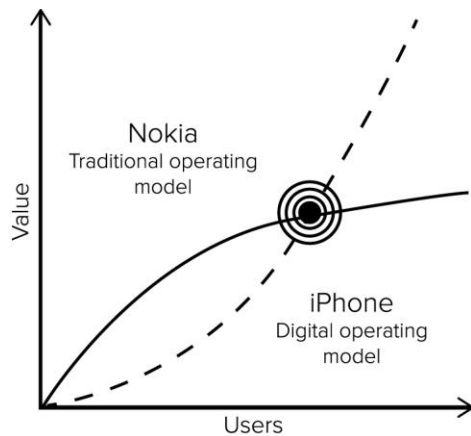
—*Forbes* cover story about Nokia, November 12, 2007, half a year after the iPhone was introduced

In chapter 6, we explored how digitizing key elements of the firm's operating model can open new strategic options and transform the way a firm creates and captures value. This chapter examines some of the broader competitive implications and explores what happens as firms featuring a digital operating model encounter and collide with more traditional firms.

A *collision* occurs when a firm with a digital operating model targets an application (or use case) that has traditionally been served by a more conventional firm (see figure 7-1). Because digital operating models are characterized by different scale, scope, and learning dynamics from those of traditional firms, collisions can completely transform industries and reshape the nature of competitive advantage.

Note that it can take quite a while for digital operating models to generate economic value that comes anywhere near the value generated by traditional operating models. This explains why executives ensconced in the traditional model have a difficult time at first believing that the digital model will ever catch up. But after the digital operating model scales beyond critical mass, the value delivered can be truly impressive, and firms operating with digital models can easily overwhelm traditional firms. The implications are increasingly felt across our entire economy.

FIGURE 7-1



The collision between a digital and a traditional firm

Take the global travel industry, where Airbnb is colliding with hotel companies like Marriott and Hilton. Airbnb serves similar needs but is built on a completely different kind of operating model. While Marriott and Hilton own and manage properties, with tens of thousands of employees in separate organizations devoted to enabling and shaping customer experiences, Airbnb's lean organization sits on top of a virtual AI factory, aggregating data and using carefully crafted algorithms to match users to its digitally tracked and managed community of property owners. And where both Marriott and Hilton are a cluster of groups and brands, each with its own siloed business units and functions equipped with their own information technology, fragmented data, and organizational structure, Airbnb's lean and agile organization sits on top of its integrated data platform, accumulating customer and process information, mining analytic insights, running rapid experiments, and producing predictive models to inform key decisions. Airbnb accumulates network and learning effects, rapidly driving scale, scope, and learning, while Marriott's growth and responsiveness are limited by its traditional operational constraints. In barely a decade, Airbnb has scaled to offer an inventory of more than 4.5 million rooms, three times as much lodging capacity as Marriott managed to accumulate in its one hundred years.

As with Amazon's supply chain or Ant Financial's credit scoring process, Airbnb moves human labor from the core of the operating model to the edge, in this case even outside the company boundaries (the hosts). Airbnb constantly mines its data to acquire new customers, identify new traveler needs, optimize experiences, and analyze risk exposure. As it does so, it accumulates even more data on hosts and travelers, and it uses artificial intelligence and machine learning to provide new insights, confirmed through frequent experimentation. Airbnb is also rapidly expanding in scope to offer a

broad variety of experiences, from concerts to flight lessons. This drives new network and learning effects and multiplies opportunities for both value creation and capture.

Airbnb isn't the only digital firm agitating the global travel market. Booking Holdings is another formidable force, whose brands—Booking.com, Kayak.com, and Priceline.com—offer 30 million listings in more than 150,000 destinations built over a lifetime slightly longer than Airbnb's. As with Airbnb, Booking is architected to enable a software- and data-centric operating model, increasing scale, scope, and learning without running into traditional operational constraints. And as with Airbnb, Booking's only real growth bottlenecks sit outside the company, in ensuring a growing inventory of travel accommodations and experiences. Booking's valuation is already twice Marriott's.

The industry is transforming in front of our eyes. In only a few years, Airbnb and Booking have dramatically increased the number of room nights sold and have catapulted into leadership positions, while increasing the bundle of services offered to consumers. Market concentration is also increasing, with M&A activity on a high boil.

Marriott has responded by merging with Starwood, aiming to exploit synergies across loyalty programs and related data assets. In a race against time, Marriott is working hard to operationalize the merger and rearchitect its operating model to remain competitive against Airbnb's and Booking's data-driven growth machines. The hotel industry is in the midst of a collision.

The Competitive Dynamics of Collision

The collision between digital and traditional travel companies shows what happens when traditional user needs are met in new ways by a different kind of operating model that digitizes some of the most critical tasks in the delivery of value. The market needs are very similar—travelers need accommodations and experiences—but unlike traditional hotel chains, Airbnb and Booking have built systems to satisfy those needs without relying on massive traditional organizations, armies of hotel managers and sales employees, and cumbersome operating processes.

Airbnb and Booking are effectively adding a software layer to the travel industry; think of it as an operating system for travel. If Marriott is the IBM mainframe company of the travel industry, Airbnb and Booking are vying to be Microsoft Windows. In doing so, they push traditional operational bottlenecks outside their organizations and remove constraints on their own scalability, scope, and learning potential.

As with computer operating systems companies, digital operating companies like Booking and Airbnb amplify the value they create by leveraging network

and learning effects. Network effects are central to their model. More demand for lodging by travelers will induce more hotels and homeowners to offer their properties online, and the more properties that are offered, the more travelers who are likely to come.

Learning effects further amplify the value delivered, as data trains machine learning algorithms to recognize patterns and improve operational decisions. Both Airbnb and Booking.com accumulate all kinds of data on user behavior, such as the types of content that a certain user will be more likely to click on, linger on, or mouse over. This data is used by an algorithm to select and prioritize the content to be steered to a user's app. As the app accumulates varied data, the learning analytics can amplify the impact of network effects as they're trained to increasingly engage the user.¹ The more data, the more refined the optimization, and the more the typical user will engage with the content.

Travel industry examples, once again, show how AI and learning and network effects can go hand in hand to build a rapidly growing value proposition for a digital operating model in a series of self-reinforcing loops. As the operating model develops more connections, it also develops increased opportunities to generate and accumulate data. The more data that is generated, the better the services the organization can provide and the more incentive there is for third parties to plug in. The better the services it provides, the more users it will attract, and the more users, the more data, and so on, in turn increasing the impact of any learning and network effects. In general, the larger the networks and the more data that is generated, the better the algorithms, and the better the algorithms, the sharper the increase of value delivered because of scale and scope.

These self-reinforcing loops in network and learning effects make a big difference in the nature of competition. The value delivered by traditional operating models becomes saturated as the organization grows. This implies that traditional operating models tend to allow for competition, enabling entrants to threaten incumbents, because the advantages of scale are significant but not insurmountable. New companies can be competitive by offering interesting, innovative solutions even on a smaller scale; think of a country inn taking room nights away from a Marriott resort. But as network and learning effects drive more of the value delivered, traditional constraints go away, and the value delivered will continue to increase, possibly at an increasing rate. If network and learning effects are strong, and if multihoming and disintermediation are rare, the viability of competitive alternatives is diminished, and markets are driven toward concentration.

As value delivered increases for digital operating models, the space left for competitors at lower scale, scope, and learning continues to shrink, making it difficult for a traditional company to sustain a profitable offering. Although

hotel companies will not cease to exist, their profits are moving to the “operating system” layer. The immense scalability of the new AI-centric “travel experience operating system” model is altering competitive dynamics, forcing Marriott, Hilton, Hyatt, and other traditional operators into the fight of their lives.

The next decade will witness an epic battle for control of the multitrillion-dollar global travel market. To get a better sense of how this battle, as well as many similar collisions, might evolve, let’s revisit the collision between traditional and digital phone providers. The story is old news, but it offers interesting insights when analyzed through our new lens.

The Classic Case

Nokia was founded in 1865 as a paper mill and eventually grew into the world leader in mobile communications. Only five years after *Forbes* featured Nokia’s industry leadership on its November 2007 cover, the company had completely collapsed. Sold to Microsoft for \$7 billion, less than one-tenth its 2007 value, Nokia’s mobile phone business was resold a couple of years later for only a few hundred million dollars. From a position of industry dominance, Nokia tumbled into irrelevance.² How could this have happened to a company that seemed to do everything right? A marvel of product innovation, design, and usability, Nokia invented most of the new features we still use on phones today, from touch screen interfaces to the first mobile internet browser. Its designs won prizes for style and usability. Its marketing organization was second to none in its relentless focus on the user. Its manufacturing processes were renowned for their high quality, low costs, and generous operating margins. In many ways, Nokia was the quintessential product company.

Nokia was architected like all other large, traditional product companies: a siloed, geographically separated, multidivisional business unit structure, dedicated product teams, and multiple R&D centers around the world. Nokia ran hundreds of simultaneous R&D projects and introduced thousands of products across more than a dozen major geographies. Its product development teams optimized integrated hardware and software features to match specific customer needs and create great designs. Supporting its product strategy was a vertically integrated manufacturing process and a dedicated and responsive supply chain. Nokia’s competitive advantage increased with the variety of differentiated models and designs—each tuned to a different geography or market segment. All this was complemented by investments in technical capabilities, patents, and branding and marketing. But as product companies often do, in order to optimize each product and tailor it to the unique needs and context of each market and organization,

Nokia sacrificed digital consistency. Even though Nokia invested heavily in the Symbian operating system, that OS was only one of several the company used. And even within Symbian products, each phone's software was fine-tuned to a different user interface design, form factor, or customer features. The developer interface, moreover, was unstable, inconsistent, and emphatically not user friendly. All this added to developers' headaches when they tried to create apps for Nokia's broad variety of models and operating system versions. Any app would have to be redesigned (by hand) for virtually every product Nokia offered. It's no surprise, then, that when Nokia opened an app store (Ovi) in 2008, the marketplace never attracted developers and never offered a critical mass of apps.

Nokia operated like any great product company would have, optimized to produce laser-focused, differentiated products. As such, it gathered no scale benefits from a standard digital foundation, no scope benefits from a successful platform ecosystem, and no learning benefits from a consistent data architecture or experimentation platform.

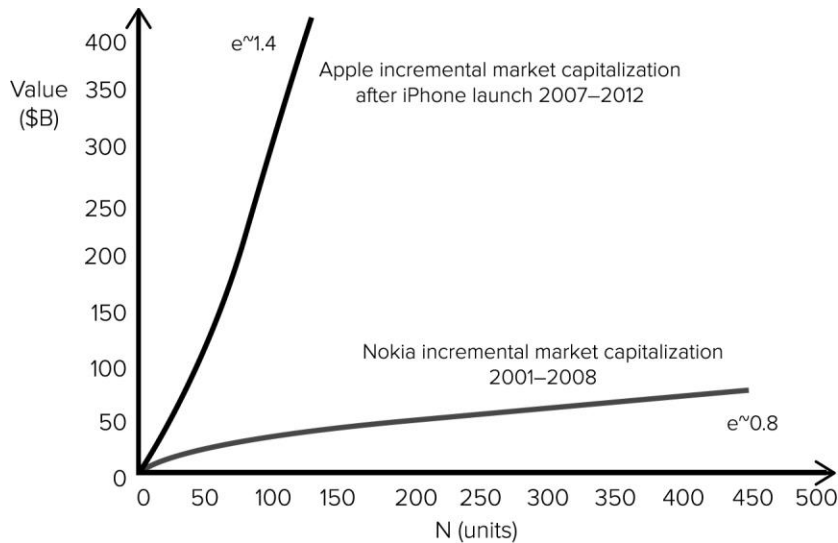
Then in 2007, Apple's iOS hit the market, rapidly followed by Google's Android. Rather than being built within a traditional, siloed product business featuring separate product units, iOS and Android phones were built on one software version, a single, consistent digital foundation. Although they could work like a phone and matched Nokia's in performance, the iPhone-iOS combination embodied a single digital platform, and soon Apple offered an elegant and consistent API, in a fashion very similar to the way PCs had been conceived since the 1980s. Android rapidly followed suit and also opened its architecture, thereby enabling a diverse set of smartphone original equipment manufacturers (OEMs).

In contrast with Nokia phones, iOS and Android phones attracted ever-expanding ecosystems of third-party app developers and service providers, complementing the core functionality embedded in the phone. Unlike Nokia's fragmented product line, the consistent iOS and Android platforms encouraged the formation of large networks of app developers, spurring intense developer excitement. The positive reinforcement loops were remarkable: the more iPhone and Android apps there were, the higher the levels of user engagement; and the higher the levels of user engagement, the greater the number of transactions and the greater the amount of data and value that flowed to developers and advertisers.

As the developer and advertiser networks reached critical mass, the value of iOS and Android increased rapidly. The slope of the value curve grew steeper as the value delivered overwhelmed the value delivered by traditional smartphones trying to serve the same customers. With millions of deployed apps, iPhone and Android took off and left Nokia's traditional, product-based

business model behind (see [figure 7-2](#)). Along with Nokia, other competitors, including BlackBerry, Sony Ericsson, and Motorola, fell off the map.

FIGURE 7-2



The Nokia and Apple value curves

Beyond displacing traditional industry leaders, the collision in smartphones dramatically changed the structure of the industry. Virtually all profits migrated from the highly competitive hardware layer to the highly concentrated software layer, capturing value through complementary revenue sources such as bundled hardware, advertising, and app download fees. The battle is not over yet, but it looks as if the final victory is likely to go to Android, already powering more than 85 percent of global smartphones. The irony is that Nokia invented and introduced many of the features we now associate with smartphones—touch screen functionality, integrated cameras, embedded search, even apps and app stores—well ahead of the iPhone introduction in 2007. Indeed, throughout the period it was losing ground to iOS and Android, Nokia was investing an impressive 8 percent to 15 percent of revenue in R&D. But iOS and Android were architected to build value in a profoundly different way. Just as Airbnb and Booking are becoming data-driven magnets for travel experience providers, both iOS and Android became magnets for app developers and advertisers. The market tipped, and Nokia and the nature of competition changed. All in all, it took less than five years. Nokia discovered that once digital network businesses reach critical mass, they can rapidly grow to dominate markets, and transform the economy.

To meet the new threat, Nokia had two options. First, it could have built its own digital operating model and competed head-on with Android and iOS. But to do this, it would have had to transition from a siloed, product-based

operating architecture to a software-optimized operating architecture—standardizing on a single consistent digital framework and adopting a standard approach to software component design, ecosystem development, and data integration. Building Symbian technology was not enough. What was required was a deep-rooted transformation, the likes of which we discussed in chapters 4 and 5.

Nokia's second option would have been to acknowledge the newfound dominance of smartphone OS companies and focus on becoming the best possible complement to the new software-based entrants. This is essentially what Samsung has done, by conceding the software battle and focusing on hardware features and components. Although not approaching the kind of value and profitability captured by iOS and Android, Samsung has survived and, to some extent, prospered. Unique to its strategy is becoming one of the industry's very few strategic suppliers of high-quality screen displays—still a highly profitable (and significant) niche. For the rest of the smartphone hardware OEMs, it's been a different story, as profits have dwindled in the brutally competitive market. However, despite the adversity, many companies still survive.

Interestingly, Nokia did not execute either option, which may explain its rapid demise. Nokia at first simply refused to change and tried to answer the threat by building more products within its existing operating architecture. But even when the failure of this approach was patently obvious, Stephen Elop, the company's CEO, refused to acknowledge Android's clear advantage and committed to the Windows mobile operating system, which already lagged far behind in market share. Without harvesting the gains of digital scale, scope, and learning, Nokia took a nosedive into oblivion.

The Pattern Repeats

The smartphone story is threatening to repeat everywhere. We have argued that Airbnb and Booking are posing a similar challenge to Marriott and Hilton. And just as cloud computing services from Amazon and Microsoft are replacing traditional software and hardware providers, marketplace platforms like Alibaba and Amazon are replacing traditional retailers. Digital, over the top (OTT) video content delivery services (think Netflix, Hulu, and Amazon Prime Video) are threatening traditional pay TV providers. New fintech companies are competing with traditional banks and insurance companies by providing data-centered financial services over the internet. Across the economy, we see traditional firms colliding with highly scalable, data-driven, software-centric operating models, leveraging networks, data, and AI to drive personalization and to expand the range of services by using digital networks to plug in to service providers. The ensuing transformation

in each of these industries is profound, and it cuts across value creation, capture, and delivery to change competitive dynamics and market structure. Let's look at a few more examples, both past and present.

Computing

The computing sector has already witnessed a number of different collisions between operating architectures, each digitizing new aspects of the industry value chain. The most impactful transformation probably took place as far back as the 1980s, when mainframe and minicomputer providers collided with personal computer firms. For the first time we saw a digital platform structure having separate, modular operating systems like CPM, DOS, and later Windows and the Mac OS. CPM fell out of favor, but the Mac OS retained an integral structure for most of its history (with Apple contributing its own applications), and Microsoft established Windows—with its hundreds and later thousands of APIs and easy-to-use Visual Studio programming tools—as the OS of choice for the industry.

Windows thus used digital interfaces to modularize and distribute the creation of software applications, thereby building a large and powerful ecosystem; at its peak there were more than six million developers whose daily occupation was building software for Windows, working for a variety of application providers. The developer ecosystem generated strong network effects, and the dominance of Windows continued for more than a decade, with Microsoft's market share in PC operating systems peaking at greater than 90 percent. In many ways, Google's increasing dominance in smartphones simply revisits the old Windows playbook, with the addition of data, AI, and the massive revenues provided by tailored advertising services. In recent years, cloud computing has led to yet another collision, in essence digitizing the process of software distribution. The cloud offers a new business and operating model for distributing a variety of computing services, with easy, network-based access to flexible computing capacity and consumption-based pricing for computing, storage, and other applications and services. The operating model for cloud computing providers is completely different from traditional software OS providers, as it hinges on the establishment of extensive data center infrastructure to efficiently deliver the services instead of either selling software in stores or deploying software on-premise in enterprises.

After losing out to Linux and other (primarily open source) alternatives, Microsoft is back in the game. Chasing Amazon Web Services, Microsoft has made great inroads in transforming business and operating models to be among the first to offer cloud services optimized for business applications. Gone are the software boxes available at Best Buy and Computer City, and

soon to be gone are the massive on-premise deployments of products like Windows Server and SQL Server; all software is now available for easy digital download, on demand, from the cloud. Not surprisingly, leadership in the industry turned over again, with Amazon (primarily through AWS) and Microsoft (after its transformation) now alternating as the world's most valuable company.

Because this industry has been dealing with collisions for so long, firms have become good at transformation. Experience is one factor, but so is the fact that the operating architecture of firms in this industry is less siloed and fragmented than you see in traditional industries. After a firm is architected as a software and data platform company, it is comparatively easier to transform it to adopt new generations of technology.

Retail

Among the first online retailers was Amazon, founded in 1994 with the emergence of the World Wide Web. Early “e-tailer” operating models like those of Amazon, drugstore.com, JD.com, or even Pets.com digitized buying transactions and moved them online. Over time, online retailers grew into actual digital retail platforms, with Amazon launching and scaling its marketplace and connecting to thousands of third-party merchants that offered unprecedented scale and scope across thousands of product categories. As described in [chapter 4](#), Amazon rearchitected its own operating model to aggregate data and share software components—designing a powerful, data-centric operating platform and driving a formidable transformation in the retail experience.

Traditional retailers stood up to the first generation of online retailers fairly well, because the transformation was relatively limited. Lacking extensive data and analytics and bottlenecked by a traditional supply chain, online retailers did not generate substantial network or learning effects. Ultimately, the likes of Pets.com and drugstore.com did not serve the unique needs of a customer any better than a traditional store. The extensive variety of goods that could be provided online was hard to navigate without personalization, and individual in-store clerks can be quite effective if well trained. The bigger threat was Amazon's rearchitected, data-centric, software-based operating model, which companies like JD.com and Wayfair emulated. The transformation consisted of more than simply moving transactions online. It called for a fundamentally different operating approach based on a data- and AI-centric unified understanding of the customer, offering a way to personalize the retail experience not only online but also offline (as we see, for example, in Amazon's Whole Foods Market acquisition). Retail supply chains became software centric, deploying labor, not in the core of the

process but on the edge (e.g., to pick odd-shaped products from shelves), an action that removed traditional bottlenecks and scale constraints. By the late 2010s, the retail apocalypse was in full force, taking down all manner of traditional players, including Toys “R” Us, Sports Authority, Nine West, and Brookstone, to name just a few.

The insight from retail is that putting a business online does not necessarily topple a traditional industry giant. The difference is having a software- and data-centric operating architecture. It was only after some of the online retailers figured this out that the industry truly transformed.³

Entertainment

The first organization to compete using a data- and software-centric operating model to collide successfully with the entertainment industry may have been Napster, which allowed people to digitize and share their music online for free—without any of the usual payments to the various players in the music industry. When it emerged in the late 1990s, it introduced music as a service. Despite its immense popularity, Napster was plagued by legal troubles and shut down in 2001. After Napster, Apple Music, Spotify, and others sparked new collisions with traditional music distribution companies, transforming the business and operating models for music distribution in the United States and beyond.

Collisions spread from music to video. Launched in 1997, RealNetworks was the first internet streaming video company.⁴ By 2000, almost all of the video streamed over the internet was in the RealNetworks format. Its business model hinged on selling server software, however, and RealNetworks suffered in competition with established software providers such as Microsoft and Apple.

Streaming services truly took off with YouTube, founded in 2005, and Netflix, which transitioned from a DVD business into a streaming service starting around 2007. YouTube and Netflix offered a more compelling value proposition for consumers as well as scalable value capture models through advertising and subscription, patterned largely on the music streaming business.

However, there is a significant difference between the Netflix and YouTube operating models, with important implications for competition. By aggregating a huge community of small content providers, YouTube accumulates important network effects and essentially dominates its market. The kinds of video streaming services provided by Netflix, on the other hand, come from a much more concentrated set of content production studios, which routinely multihome and offer their content across various delivery platforms. Although Netflix’s data and learning advantages are important,

they do not add up to the kind of advantage enjoyed by YouTube at scale. This has enabled a number of companies to sustain competitive offerings, from Hulu to Amazon. Lacking strong network effects, each of these providers is attempting to differentiate itself by accessing unique content through special studio relationships and vertical integration. Digital firms now have huge content production budgets and are challenging traditional providers in most global markets.

As a group, Google, Netflix, Apple, and Amazon are colliding with traditional cable and satellite television providers, providing OTT internet-based video content distribution platforms that have rapidly scaled to hundreds of millions of users globally. Despite differences in the network effects accumulated, each of these firms competes on a data-centric operating model, driving extensive customization and personalization to cater each viewing experience to the needs of individual users. On alert after the devastation in music and retail, traditional media companies are scrambling to react, merging with content and internet service providers to spark transformation and rearchitecting operations around a digital core. Comcast and Disney have demonstrated important progress, from the development of the X1 platform to ESPN streaming services.

The transformation of entertainment reveals other interesting patterns. First, the original innovator in a given industry does not always win; Napster is long gone. Deploying a digital operating model is not enough. For a collision to threaten established players, the innovator needs an effective business model as well. Additionally, as they compete with traditional companies, digital firms compete with each other. As they do so, they may emerge as focused competitors like Netflix, or they may leverage synergies in assets and capabilities across industries, like Amazon and Apple. The winners and level of concentration in each market will be shaped by the resulting economies of scale, scope, and learning.

Automotive

Cars are becoming increasingly connected and digital, and this increased connectivity and functionality is threatening the traditional operating models of automobile companies. At stake is the enormous value of connecting to consumers while in transit—for example, during the work commute, which, on average, takes around one hour a day in the United States. The value of one hour of consumer access is high—hundreds of billions of dollars in the United States alone.

Exploiting opportunities to extract economic value from a moving, connected car will require a digital, data-centric operating model that delivers a marketplace of on-demand services or highly targeted ads that will be

embedded in the cars themselves, through various screens or audio aimed at drivers and passengers. Car-sharing services like Uber, Lyft, and DiDi are starting to show the way, but the best opportunity lies in autonomous driving systems. When consumers no longer need to pay attention to driving, they'll want entertainment and social interactions—turning the car into a large smartphone on wheels. It's no surprise, then, that new and old companies are engaged in a fight for the incremental value created and captured.

Alphabet is first in line. Already at scale from its mobile business, Android is ready to shape automotive user behavior and value capture for its parent company. Google Maps and advertising networks are also already at scale and ready to create relevant local ads pinpointed to the car's location. The next step is literally driving users to commercial opportunities. Auto manufacturers, pressured by consumer demands, have given hub firms dashboard screen access in many cars, directly integrating their services into the driving experience. Adding to these already massive opportunities, Alphabet subsidiary Waymo is developing a driverless car as a service business, which, by itself, could one day earn hundreds of billions of dollars in revenue.

These changes will transform the industry. As the trends continue, transportation will become less about car ownership and experience, and more about the convenience and services offered by automobiles as they drive passengers around. Sure, some people will still want cars they can actually drive, but differentiation will lessen, and most car hardware might well become increasingly commoditized, just as with most Android OEMs. Just as we saw in other examples, the effects of transformation in the auto industry won't be limited to auto manufacturers; they will upend a range of connected sectors, including insurance companies, repair and maintenance providers, road and construction companies, law enforcement, and infrastructure providers, as the digital dominos continue to fall. Even governments will be affected, because many local, state, and federal authorities rely on various forms of automotive taxes.

As the Nokia saga suggests, the auto manufacturers' core business will increasingly be commoditized as a more concentrated software layer emerges. Revenues and margins will erode as demand saturates and car utilization increases. As differentiation moves from hardware to software and networks, now largely outside the manufacturers' control, price premiums will plummet.

What can traditional auto manufacturers do? As with Nokia, they appear to have two options: either challenge hub firms like Alphabet and Apple, or work with them and become their chosen, best suppliers. Both strategies come with challenges. The first entails competing with the likes of Android and iOS, which are already at scale, and includes crucial services like maps

and advertising platforms. The second involves resisting the commoditization of automotive hardware and its components as functionality and market power move to the software layer.

As the traditional automotive business appears to head toward commodity status, some auto manufacturers are attempting to participate in the emerging software and services layers of the automotive stack. Indeed, some automakers are preparing for a pay-per-use model for car usage, and several manufacturers have already acquired or partnered with major car-as-service providers, as with GM's investment in Lyft or Daimler's acquisition of car2go. Several manufacturers have also invested in their own driverless research or have partnered with external providers. The key issue is whether they will be able to gather enough scale, scope, and learning advantages to compete with the pros.

Beyond investing in digital transformation and experimenting with new, service-based business and operating models, automakers may need to play the way the digital hubs do. And to reach the scale required to be competitive, once fiercely competitive automotive companies will need to rearchitect their operating models and even join forces to aggregate enough scale.

HERE, a precision map and location service provider, is an interesting example. HERE has its roots in Navteq, one of the early online mapping companies, acquired first by Nokia and more recently by a consortium of Volkswagen, BMW, and Daimler. Providing a sophisticated set of tools and APIs to enable third-party developers to develop location-based ads and other services, HERE is an attempt by traditional auto manufacturers to work together to assemble a "federated" platform. In doing so, HERE neutralizes a potential competitive bottleneck and counterbalances clear threats from Google and Apple. The consortium could play a significant role in preventing automotive value capture from tipping completely toward existing digital firms.

The next decade will entail major changes and transformation for the automotive sector. Traditional manufacturers should not underestimate the competitive skill and scale, scope, and learning advantages exhibited by the digital firms that are entering the space. They have played this game before and obviously understand the new shape of competition.

Where We Are Heading

We are witnessing a new generation of digital operating models transforming the economics and nature of service delivery. Software, along with data- and AI-centric architectures, is removing traditional operational constraints and enabling a new generation of business models that cut across industries. This

is transforming competition, and we already see evidence of a more concentrated, winner-take-all world emerging in some traditional markets. And as collisions multiply across the economy, different industries become increasingly connected to each other, through the new, ubiquitous digital fabric. Our entire economy is starting to resemble a vast, highly connected network coalescing around a small number of digital superpowers.

A generation of hub firms has emerged—the likes of Apple, Alphabet/Google, Amazon, Baidu, Facebook, Microsoft, Tencent, and Alibaba, including many of the examples covered in this book. Beyond challenging some traditional competitors, hub firms are enabled by their operating models to occupy increasingly central positions in our economy, reaching across to connect and orchestrate traditionally disparate industries. While creating real value for users, these companies also capture a large and expanding share of the value generated, and they're shaping our collective future.

Beyond influencing individual markets, hub firms are poised to create and control essential connections in key networks. The Android operating system is forming a competitive bottleneck well beyond the phone industry, owning access to billions of consumers that other product and service providers want to reach. Amazon's and Alibaba's marketplaces connect vast numbers of users with vast numbers of retailers and manufacturers. Tencent's WeChat messaging platform aggregates a billion global users and provides a critical source of consumer access for businesses offering online banking, entertainment, transportation, and other services. Alibaba is connecting e-commerce transactions with credit scoring, investment management, and loans, all at an unprecedented scale.

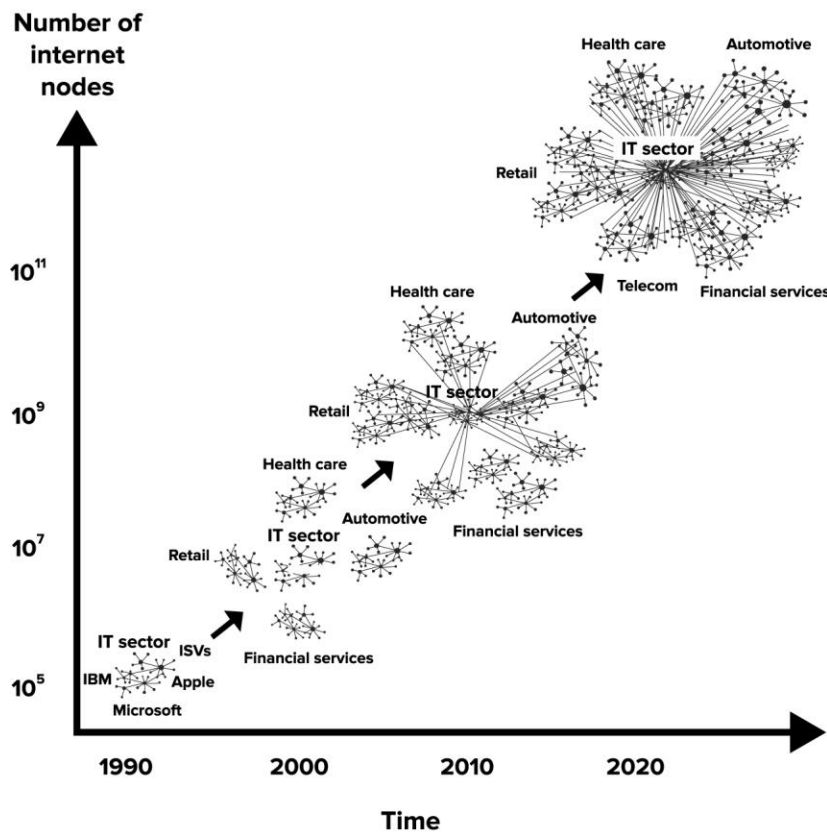
The more users join these networks, the more attractive (and even mandatory) it becomes for enterprises to offer their products and services through them. By driving increasing returns to scale, scope, and learning, these digital superpowers can control crucial competitive bottlenecks, extract disproportionate value, and tip the global competitive balance, as suggested in [figure 7-3](#). The implications, as we are all witnessing, go well beyond the economy.

The speed with which traditional processes are being replaced by digital technology is increasing at what is starting to feel like an exponential rate. The introduction of software platforms provided an initial impulse, but the technologies are becoming sophisticated enough to rapidly transcend relatively simple software applications. The impact of data, analytics, and AI is just ramping up and has ways to go. And as digital technology increasingly collides with disparate aspects of our economy and society, Nokia's fate is threatening industries as diverse as media and banking, automotive and travel. After a hundred years of history, companies like Marriott and Hilton

are investing in driving major transformation, integrating disparate data assets, developing capabilities in analytics and AI, and working hard to rearchitect their traditional operating models.

Beyond shaping the fate of leading firms, the impact of these collisions is being felt across our entire economy and into our social and political system. As disparate industries increasingly consolidate into one giant network, the concentration of value and information not only creates opportunity but also introduces new problems. From the erosion of consumer privacy to the emergence of an increasing variety of cyber threats, and from disinformation campaigns to economic disparity, the spread of digital operating models is causing a range of new threats.

FIGURE 7-3



The evolution of the modern economy

Managers will have their work cut out for them, as they reflect on their evolving role in our increasingly digital economy. The next chapter focuses on some of these considerations.